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Bayesian Inference for Stochastic Volatility Models in Financial Markets

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Abstract

This thesis provides a theoretical study of Bayesian inference for stochastic volatility models in financial markets. The focus is on the mathematical structure of Bayesian inference, how it addresses the latent-state estimation problem in the stochastic volatility models, and what does it change relative to classical conditional volatility models. The analysis begins by reviewing the main stylized facts of financial returns. ARCH and GARCH models are introduced as observation-driven benchmarks, where the conditional variance is updated recursively from past returns and past variances. Stochastic volatility models are then presented as latent state-space models, in which the conditional variance is determined by an unobserved stochastic process. This distinction creates the essential inferential problem of the thesis: the likelihood of the observed returns depends on a high-dimensional latent path that cannot be integrated out analytically in closed form. Bayesian inference overcomes this limitation by treating the latent volatility path as part of the inferential object, alongside with the model parameters. The thesis studies the resulting posterior distribution, marginal posterior quantities, prior restrictions, latent-state uncertainty, and the analytical intractability that motivates simulation-based methods. In particular, data augmentation, Markov chain Monte Carlo, Gibbs sampling, and Metropolis–Hastings are discussed as computational tools for approximating posterior distributions in Bayesian stochastic volatility models. The main result is that the value of Bayesian inference lies in its coherent treatment of parameter uncertainty, latent volatility uncertainty, and predictive uncertainty within a unified probabilistic framework.

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1 Introduction

1.1 Financial volatility and unobserved risk

Financial markets are characterized by uncertainty. Asset prices incorporate information about future cash flows, liquidity conditions, macroeconomic risk, and investor expectations, but these determinants are not known with certainty at the time at which the portfolio is made. For a traded asset with price process $\{P_t\}_{t \geq 0}$, the basic object of analysis is usually the continuously compounded return

$$r_t = \log P_t - \log P_{t-1}. \quad (1)$$

Even when the conditional mean of r_t is small at high frequencies, the conditional dispersion of r_t is economically central. Volatility is therefore not only a statistical description of return variability, but also an important measure of financial risk.

The empirical literature on financial returns shows that the actual returns depart from the classical model of independent and identically distributed Gaussian observations. Mandelbrot (1963) [15] documented large price changes and heavy-tailed behaviour in speculative prices, while Fama (1965) [8] provided early evidence on the distributional and dependence properties of stock-market prices. Later summaries describe these features as stylized facts: return distributions are leptokurtic, meaning that extreme observations occur more often than under a normal distribution, and large absolute returns tend to be followed by further large absolute returns, a phenomenon known as volatility clustering (Cont, 2001 [5]). These properties imply that the variance of returns is not constant over time and that periods of market calm and turmoil alternate in a persistent way (Tsay, 2010 [22]).

Volatility is extremely relevant because many financial decisions depend on the conditional distribution of future returns. As an example, in risk management, measures such as Value-at-Risk and expected shortfall are sensitive to the scale and tail behaviour of the predictive distribution. The practical problem is that volatility is not seen in the same way as prices or returns. The general econometrician can observe P_t or r_t , but the underlying level of risk that

generated the return remains hidden.

This unobservability is the starting point of the thesis. Classical volatility models and stochastic-volatility models both attempt to describe time-varying risk, but they differ in how they treat the unobserved variance. ARCH and GARCH models construct volatility as a deterministic conditional variance, updated from past returns and past variances, while stochastic volatility models treat volatility as a latent stochastic process.

1.2 Motivation for stochastic volatility models

The autoregressive conditional heteroskedasticity model of Engle (1982) [7] introduced the idea that the conditional variance of a time series can depend on past squared innovations. Bollerslev (1986) [2] generalized this structure through the GARCH model, in which the conditional variance follows a recursion such as

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \omega > 0, \quad \alpha, \beta \geq 0. \quad (2)$$

Here, σ_t^2 denotes the conditional variance of returns at time t , r_{t-1}^2 represents the squared return from the previous period, and σ_{t-1}^2 is the previous conditional variance. This class of models captures volatility clustering through deterministic updating: a large past shock increases the next conditional variance, while a high value of β implies that volatility decays slowly back toward its long-run level. ARCH and GARCH models therefore provide a natural benchmark for conditional heteroskedasticity.

In stochastic volatility models, instead of assuming that variance is fully determined by past returns and past variances, a latent state is introduced, that evolves according to its own probability law. A basic discrete-time SV model can be written as

$$r_t = \exp(h_t/2)\varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad (3)$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2), \quad (4)$$

where h_t is latent log-volatility. The parameter μ describes the long-run level of log-volatility,

ϕ measures volatility persistence, and σ_η controls the size of volatility shocks.

The same feature that makes this model flexible also makes inference significantly more difficult. In a GARCH model, the conditional variance can be computed recursively once the parameters and past observations are known. On the other hand, in a stochastic volatility model, one observes $r_{1:T}$, but not the latent path $h_{1:T}$. As a result, the likelihood of the observed return sequence must average over all possible volatility paths. In compact notation, for parameters $\theta = (\mu, \phi, \sigma_\eta^2)$,

$$p(r_{1:T} | \theta) = \int p(r_{1:T} | h_{1:T}, \theta) p(h_{1:T} | \theta) dh_{1:T}. \quad (5)$$

This integral is high-dimensional because the number of latent volatility states $h_{1:T}$ increases with the sample size. Therefore, stochastic volatility models are theoretically appealing precisely because they treat volatility as hidden and random, but this also makes their estimation substantially more demanding.

1.3 Bayesian inference and research question

Bayesian inference provides a natural framework for SV models because it treats the unknown quantities probabilistically. This includes not only the static parameters θ , but also the entire latent path $h_{1:T}$. Bayesian analysis combines the likelihood of observed returns, the state dynamics, and prior distributions into the joint posterior distribution

$$p(\theta, h_{1:T} | r_{1:T}) \propto p(r_{1:T} | h_{1:T}, \theta) p(h_{1:T} | \theta) p(\theta). \quad (6)$$

This posterior is the object from which multiple metrics can be derived, such as parameter estimates, latent-volatility estimates, and risk measures.

Early Bayesian contributions showed that the latent structure of stochastic volatility models could be handled through simulation-based inference. Jacquier, Polson and Rossi (1994) [13] developed one of the first Bayesian approaches to this problem by treating the unobserved volatility path as part of the set of unknown quantities to be estimated. Kim, Shephard and Chib (1998) [14] later provided influential likelihood-based and simulation-based methods

that made stochastic volatility models more practical for econometric applications. These contributions are extremely relevant because they transformed SV modelling from a theoretical structure into an operational econometric framework.

The central research question of this thesis is:

How does Bayesian inference address the latent-state estimation problem in stochastic volatility models, and what does it add relative to ARCH/GARCH models?

It is therefore not the purpose of this thesis to claim that Bayesian SV models are always better than ARCH or GARCH models. Instead, the goal is to explain the distinct nature of their inference procedures. Conditional heteroskedasticity models provide an analytical recursion for the variance, while the Bayesian SV ones provide a probabilistic way to learn about the unobservable volatility process. The contribution of this thesis is to clarify why the latent-state structure of stochastic volatility models naturally leads to a Bayesian formulation. In particular, to show how Bayesian inference provides a coherent framework for estimating both the model parameters and the unobserved volatility path, while also quantifying uncertainty about them.

2 Financial Time Series and Classical Volatility Models

2.1 Returns and log-returns

Let P_t denote the observed price of a financial asset at time t , where $t = 0, 1, \dots, T$. The price series is therefore:

$$P_0, P_1, \dots, P_T, \quad (7)$$

with $P_t > 0$ for all t . Although asset prices are the directly observed market quantities, volatility models are usually formulated in terms of returns. The main reason is that price levels are often non-stationary, meaning they might display trends or long-run movements that make their statistical properties unstable over time. Returns remove much of this level effect and are therefore more suitable for time-series modelling (Campbell, Lo and MacKinlay, 1997 [3]; Tsay, 2010 [22]).

The simple return from time $t - 1$ to time t is defined as

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1. \quad (8)$$

It measures the relative change of the asset price over one period. From this previous formulation one can define the gross return

$$1 + R_t = \frac{P_t}{P_{t-1}}, \quad (9)$$

which expresses the gain/loss as a percentage of the previous price.

In financial econometrics, it is also common to work with continuously compounded returns (log-returns), defined by:

$$r_t = \log P_t - \log P_{t-1} = \log \left(\frac{P_t}{P_{t-1}} \right). \quad (10)$$

Log-returns are convenient because they are additive over time:

$$\log \left(\frac{P_t}{P_{t-k}} \right) = \sum_{j=0}^{k-1} (\log P_{t-j} - \log P_{t-j-1}) = \sum_{j=0}^{k-1} r_{t-j}. \quad (11)$$

Moreover, they are approximately equal to simple returns when returns are small. Since

$$r_t = \log(1 + R_t), \quad (12)$$

we can use the Taylor expansion of $\log(1 + x)$ around zero and setting $x = R_t$:

$$r_t = R_t - \frac{R_t^2}{2} + \frac{R_t^3}{3} - \frac{R_t^4}{4} + \dots. \quad (13)$$

For small returns, the higher-order terms are negligible, so

$$r_t \approx R_t. \quad (14)$$

Finally, log-returns are very useful in both discrete-time and continuous-time models, because they transform multiplicative price changes into additive increments.

For these reasons, the basic object of volatility modelling is usually not P_t , but the return process $\{r_t\}_{t=1}^T$. The main statistical question is not only how the conditional mean of r_t evolves, but more importantly how its conditional variance changes over time. In financial applications, this conditional variance is interpreted as the volatility expected at time t given the information available before observing r_t (Tsay, 2010 [22]).

2.2 Stylized facts of financial returns

A simple benchmark model for returns would assume that

$$r_t \sim N(\mu, \sigma^2), \quad (15)$$

where both the mean μ and the variance σ^2 are constant over time. In this setting, returns are Gaussian, meaning the probability of extreme observations is determined by the normal

distribution, and volatility is constant over time. However, empirical evidence shows that financial returns usually violate this simple description. The main regularities are known as stylized facts, meaning empirical properties observed across many different assets and markets (Mandelbrot, 1963 [15]; Fama, 1965 [8]).

A first stylized fact is that raw returns often show weak linear autocorrelation. For many financial return series,

$$\text{Corr}(r_t, r_{t-k}) \approx 0 \quad (16)$$

for several lags k . This does not mean that returns are independent, but it suggests that the conditional mean is difficult to predict using only past returns.

A second stylized fact is the presence of heavy tails. Financial returns tend to produce large positive and negative observations more frequently than the normal distribution predicts. One way to express this property is through kurtosis:

$$\kappa = \frac{\mathbb{E}[(r_t - \mu)^4]}{\sigma^4}, \quad (17)$$

where μ and σ^2 are the mean and variance of r_t . For a normal distribution, $\kappa = 3$, but empirical return distributions often display $\kappa > 3$, which is referred to as excess kurtosis. This implies that a Gaussian model with constant variance may underestimate the probability of extreme losses or gains.

A third and central stylized fact is volatility clustering. What it means is that large returns in absolute value tend to be followed by large returns in absolute value, and small returns tend to be followed by small returns. A typical way to express this property is by conditioning that the squared returns are positively correlated:

$$\text{Corr}(r_t^2, r_{t-k}^2) > 0 \quad (18)$$

for a range of lags k . Since squared returns r_t^2 are commonly used as a rough proxy for volatility, this condition implies persistence in the scale of return variability.

2.3 Volatility clustering

To formalize volatility modelling, it is necessary to distinguish between dependence in returns and dependence in volatility. Let $\mathcal{F}_{t-1}$ denote the information available immediately before time t . In a simple financial time-series setting, $\mathcal{F}_{t-1}$ contains past prices, past returns, and any other information observable up to time $t - 1$. Mathematically, it may be represented as a filtration, for example

$$\mathcal{F}_{t-1} = \sigma(r_{t-1}, r_{t-2}, \dots), \quad (19)$$

where $\sigma(\cdot)$ denotes the sigma-algebra generated by the past return sequence.

A general return decomposition can be written as

$$r_t = \mu_t + \varepsilon_t, \quad (20)$$

where $\mu_t = \mathbb{E}(r_t \mid \mathcal{F}_{t-1})$ is the conditional mean and ε_t is the innovation, namely the unpredictable part of the return given the information set. By construction,

$$\mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0. \quad (21)$$

The conditional variance of returns is then defined as

$$\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1}) = \text{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}), \quad (22)$$

with the latter transformation due to the fact that μ_t is predictable at time $t - 1$, being based on $\mathcal{F}_{t-1}$, so it behaves like a constant and does not change the variance.

From now on this becomes the central definition of volatility used throughout the thesis. Volatility is not simply the unconditional variance of the full return distribution, but rather the variance expected for the next return, conditional on the information available at that time.

A significant result coming from these expressions of volatility is conditional heteroskedasticity, which is the formal concept behind ARCH and GARCH models. This property means the

conditional variance depends on time and on the information set:

$$\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2. \quad (23)$$

2.4 ARCH/GARCH as classical benchmarks

ARCH and GARCH models are classical approaches in applied financial econometrics for modelling time-varying conditional variance. Since short-horizon returns often display weak linear predictability, these models usually place limited emphasis on the conditional mean and focus instead on the dynamics of volatility.

2.4.1 ARCH model

The ARCH model, introduced by Engle (1982) [7], explains current volatility as a function of past squared shocks. Consider the return equation

$$r_t = \mu + \varepsilon_t, \quad (24)$$

where the innovation is written as

$$\varepsilon_t = \sigma_t z_t, \quad (25)$$

and $\{z_t\}$ is a sequence of standardized innovations satisfying $\mathbb{E}(z_t) = 0$ and $\text{Var}(z_t) = 1$. The ARCH(q) variance equation is

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2. \quad (26)$$

Since the variance equation depends on squared shocks, both large positive and large negative innovations increase volatility.

The usual parameter restrictions are

$$\omega > 0, \quad \alpha_i \geq 0, \quad i = 1, \dots, q. \quad (27)$$

These ensure that σ_t^2 is non-negative for every possible history of shocks and each lagged shock contributes positively to volatility.

The unconditional variance of ε_t can be derived from the ARCH(q) equation by using the fact that the process is stationary and all the lagged shocks have the same unconditional variance and is expressed as

$$\text{Var}(\varepsilon_t) = \frac{\omega}{1 - \sum_{i=1}^q \alpha_i}. \quad (28)$$

For it to be finite and positive, it is necessary that

$$1 - \sum_{i=1}^q \alpha_i > 0, \quad (29)$$

which gives the stationarity condition

$$\sum_{i=1}^q \alpha_i < 1. \quad (30)$$

2.4.2 GARCH model

The GARCH model of Bollerslev (1986) [2] extends ARCH by including lagged conditional variances. The GARCH(p, q) model is

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2. \quad (31)$$

In addition to the effect of past shocks, the terms $\beta_j \sigma_{t-j}^2$ measure the persistence of past conditional variance. The most common specification is GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (32)$$

Here, ω is a constant component, α measures the reaction of volatility to new information, and β measures volatility persistence. The sum $(\alpha + \beta)$ is often interpreted as the overall persistence of volatility.

For the GARCH(1,1) model, the standard positivity restrictions are

$$\omega > 0, \quad \alpha \geq 0, \quad \beta \geq 0. \quad (33)$$

With the unconditional variance now being

$$\text{Var}(\varepsilon_t) = \frac{\omega}{1 - \alpha - \beta}. \quad (34)$$

This formula clearly shows that the long-run variance depends on both the constant term and the persistence of the volatility recursion.

GARCH models are important because they were proven empirically effective at capturing volatility clustering, while their recursive structure makes estimation relatively tractable. Under Gaussian innovations, the conditional distribution is

$$r_t \mid \mathcal{F}_{t-1} \sim N(\mu, \sigma_t^2), \quad (35)$$

and the conditional log-likelihood for parameters θ can be written as

$$\ell_T(\theta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log(\sigma_t^2) + \frac{(r_t - \mu)^2}{\sigma_t^2} \right]. \quad (36)$$

The likelihood is evaluated recursively because σ_t^2 depends on past residuals and past conditional variances. In this setting, maximizing this conditional likelihood gives the classical maximum likelihood estimator (MLE).

2.5 Limitations of deterministic conditional variance models

The main limitation of standard GARCH models is that volatility is deterministic, conditional on past information and parameters. In the GARCH(1,1) model,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (37)$$

once ω , α , β , past shocks, and the previous conditional variance are known, σ_t^2 becomes fixed. There is no separate random shock driving volatility itself. This is central to the motivation for stochastic volatility models, which introduce an additional state equation for the evolution of volatility.

A second limitation of the basic GARCH model is its symmetric response to shocks. Since the variance equation depends on ε_{t-1}^2 , a positive and a negative shock of the same magnitude have the same effect on future volatility. This is restrictive in financial markets where negative returns often increase future volatility more than positive returns.

3 Stochastic Volatility Models

3.1 The basic discrete-time SV model

The standard discrete-time stochastic volatility model for a demeaned return series $\{r_t\}_{t=1}^T$ is

$$r_t = \exp(h_t/2)\varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad (38)$$

where h_t denotes latent log-volatility. In particular,

$$h_t = \log(\sigma_t^2), \quad (39)$$

or equivalently,

$$\sigma_t^2 = \exp(h_t). \quad (40)$$

The exponential transformation ensures that the conditional variance is positive for all values of h_t . Conditional on h_t , the return has $\mathbb{E}(r_t \mid h_t) = 0$ and

$$\text{Var}(r_t \mid h_t) = \exp(h_t). \quad (41)$$

Thus, the latent state h_t determines the conditional variance of the observed return.

The log-volatility process is usually specified as a stationary autoregressive process of order one:

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \quad (42)$$

The two shocks ε_t and η_t are assumed independent in the basic model. The parameter vector is

$$\theta = (\mu, \phi, \sigma_\eta^2). \quad (43)$$

The parameter μ is the long-run mean of log-volatility, ϕ is the persistence parameter, and σ_η^2 is the variance of volatility shocks.

The stationarity condition for the latent AR(1) process is

$$|\phi| < 1. \quad (44)$$

What this means in practice is that h_t will fluctuate around a stable long-run level rather than drifting away forever. Its unconditional distribution is

$$h_t \sim N \left(\mu, \frac{\sigma_\eta^2}{1 - \phi^2} \right). \quad (45)$$

When $\phi \approx 1$, shocks to volatility are persistent and high-volatility periods tend to last for many observations. This is one of the main ways in which the model reproduces volatility clustering.

Equations (38) and (42) are introduced to separate the two sources of uncertainty. The shock ε_t affects the return directly, while the shock η_t affects the volatility process. This separation is a key difference from the basic GARCH recursion, allowing returns to be unpredictable in their mean while their conditional variance follows a persistent hidden process (Taylor, 1986 [20]; Jacquier, Polson and Rossi, 1994 [13]).

3.2 State-space representation

The SV model can be expressed as a state-space model. A state-space model has an observation equation, which describes the distribution of the observed data conditional on the latent state, and a state equation, which describes the evolution of the latent state itself.

From the basic SV model, the observation equation can be written as

$$r_t \mid h_t, \theta \sim N(0, \exp(h_t)), \quad (46)$$

where the mean $\mu_r = 0$ because the returns are demeaned, while the state equation is

$$h_t \mid h_{t-1}, \theta \sim N(\mu + \phi(h_{t-1} - \mu), \sigma_\eta^2). \quad (47)$$

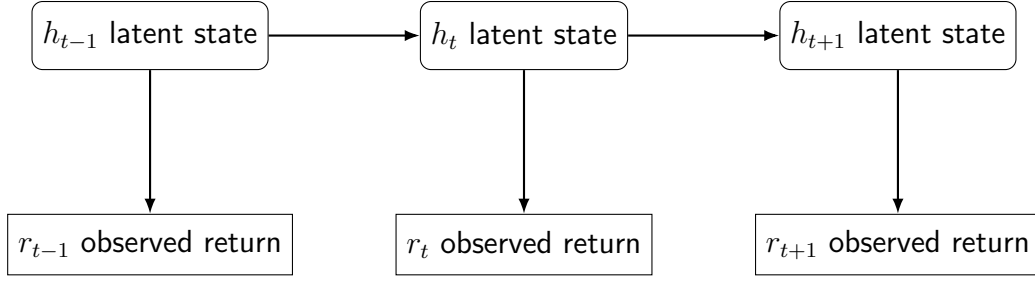


Figure 1: State-space structure of the basic stochastic volatility model

The observed sequence is

$$r_{1:T} = (r_1, \dots, r_T), \quad (48)$$

whereas the latent volatility path is

$$h_{1:T} = (h_1, \dots, h_T). \quad (49)$$

Inference requires learning both the fixed parameter vector θ and the entire hidden path $h_{1:T}$.

The probabilistic structure of the SV model is expressed by the joint density

$$p(r_{1:T}, h_{1:T} \mid \theta). \quad (50)$$

By the chain rule, this joint density can be decomposed as

$$p(r_{1:T}, h_{1:T} \mid \theta) = p(h_{1:T} \mid \theta) p(r_{1:T} \mid h_{1:T}, \theta). \quad (51)$$

The latent log-volatility process is Markovian, since the distribution of h_t depends on the past only through h_{t-1} . Therefore,

$$p(h_{1:T} \mid \theta) = p(h_1 \mid \theta) \prod_{t=2}^T p(h_t \mid h_{t-1}, \theta). \quad (52)$$

Moreover, conditional on the latent states $h_{1:T}$, the observed returns are conditionally independent. Hence,

$$p(r_{1:T} \mid h_{1:T}, \theta) = \prod_{t=1}^T p(r_t \mid h_t, \theta). \quad (53)$$

Combining these two expressions gives

$$p(r_{1:T}, h_{1:T} | \theta) = p(h_1 | \theta) \prod_{t=2}^T p(h_t | h_{t-1}, \theta) \prod_{t=1}^T p(r_t | h_t, \theta). \quad (54)$$

This factorization gives the complete-data density, involving both the observed returns and the latent volatility path. The term $p(h_1 | \theta)$ specifies the initial distribution of the latent state. For a stationary initialization, assuming $|\phi| < 1$, the first state density is

$$h_1 | \theta \sim N\left(\mu, \frac{\sigma_\eta^2}{1 - \phi^2}\right). \quad (55)$$

Therefore,

$$p(h_1 | \theta) = \frac{1}{\sqrt{2\pi\sigma_\eta^2/(1 - \phi^2)}} \exp\left(-\frac{(h_1 - \mu)^2}{2\sigma_\eta^2/(1 - \phi^2)}\right). \quad (56)$$

Since $h_{1:T}$ is unobserved, the likelihood for θ must be obtained by integrating the joint density over all possible volatility paths. The likelihood of the observed returns is therefore

$$p(r_{1:T} | \theta) = \int_{\mathbb{R}^T} p(r_{1:T}, h_{1:T} | \theta) dh_{1:T}. \quad (57)$$

Using the factorization in equation (54), this becomes

$$p(r_{1:T} | \theta) = \int_{\mathbb{R}^T} p(h_1 | \theta) \prod_{t=2}^T p(h_t | h_{t-1}, \theta) \prod_{t=1}^T p(r_t | h_t, \theta) dh_1 \cdots dh_T. \quad (58)$$

This integral is high-dimensional because it involves the entire latent volatility path $h_{1:T}$. Unlike in GARCH models, the volatility sequence cannot be computed deterministically from past returns. Instead, it must be inferred as an unobserved random process, this being the main reason why SV models are computationally more demanding (Jacquier, Polson and Rossi, 1994 [13]; Kim, Shephard and Chib, 1998 [14]; Durbin and Koopman, 2012 [6]).

3.3 Distributional properties

Conditional on the latent log-volatility h_t , the basic SV model is Gaussian:

$$r_t \mid h_t, \theta \sim N(0, \exp(h_t)). \quad (59)$$

However, the unconditional distribution of r_t is not Gaussian in general because h_t is random.

It can be written as

$$p(r_t) = \int p(r_t \mid h_t) p(h_t) dh_t. \quad (60)$$

Thus, the return distribution is a mixture of normal distributions with different variances. This mixture structure can generate heavier tails than a single normal distribution with constant variance, even though returns are conditionally Gaussian.

The model also reproduces volatility clustering. Since h_t follows an AR(1) process, a high value of h_t is likely to be followed by a high value of h_{t+1} . Consequently, the conditional variance $\exp(h_t)$ is persistent. This generates periods of high and low volatility without requiring strong linear predictability in returns themselves.

3.4 The latent-state inference problem

The theoretical appeal of SV models comes with a difficult estimation problem. The volatility states $h_{1:T}$ are not observed, so they must be inferred from returns. The likelihood of parameters requires marginalizing over the latent path:

$$p(r_{1:T} \mid \theta) = \int p(r_{1:T}, h_{1:T} \mid \theta) dh_{1:T}. \quad (61)$$

Using the factorization in (54), this becomes an integral over T latent variables. As T grows, the dimension of the integration problem grows with the sample size.

The inference problem therefore has two components. The parameter vector $\theta = (\mu, \phi, \sigma_\eta^2)$ must be estimated and the latent log-volatility path $h_{1:T} = (h_1, \dots, h_T)$ must be inferred from the observed returns.

Bayesian inference addresses this issue by treating both the parameters and the latent volatility path as unknown. Instead of first estimating a single volatility path and then estimating parameters conditionally on that path, the Bayesian approach constructs the joint posterior distribution

$$p(\theta, h_{1:T} \mid r_{1:T}). \quad (62)$$

This posterior distribution summarizes uncertainty about both the fixed parameters and the hidden volatility states (Jacquier, Polson and Rossi, 1994 [13]; Kim, Shephard and Chib, 1998 [14]).

4 Bayesian Inference for Latent Volatility Models

4.1 Bayesian formulation: prior, likelihood, and posterior

In a standard parametric model, Bayesian inference starts from Bayes' Rule. Let y denote the observed data and let θ denote the unknown parameter vector. The posterior distribution of θ is given by

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{p(y)}, \quad (63)$$

where $p(\theta)$ is the prior distribution, $p(y | \theta)$ is the likelihood function, and $p(y)$ is the marginal density of the data. Since $p(y)$ does not depend on θ , the posterior is often written as

$$p(\theta | y) \propto p(y | \theta)p(\theta). \quad (64)$$

This proportionality is the key to understanding the representation required for Bayesian analysis, where the posterior distribution incorporates prior knowledge through $p(\theta)$ and sample knowledge through $p(y|\theta)$.

In stochastic volatility models, the same principle applies, but the inferential problem is more complex because volatility is not directly observed. The unknown quantities are not only the parameter vector $\theta = (\mu, \phi, \sigma_\eta^2)$, but also the entire latent volatility path $h_{1:T}$.

Therefore, Bayesian inference targets the joint posterior distribution of both the parameters and the latent states, conditional on the observed return sequence $r_{1:T}$:

$$p(\theta, h_{1:T} | r_{1:T}) \propto p(r_{1:T} | h_{1:T}, \theta)p(h_{1:T} | \theta)p(\theta). \quad (65)$$

The first term on the right-hand side is the likelihood conditional on the latent volatility path. It is important to notice that the observations are independent conditional on $h_{1:T}$, so that

$$p(r_{1:T} | h_{1:T}, \theta) = \prod_{t=1}^T p(r_t | h_t, \theta). \quad (66)$$

For the basic Gaussian stochastic volatility model, where

$$r_t \mid h_t \sim N(0, \exp(h_t)), \quad (67)$$

this likelihood becomes

$$p(r_{1:T} \mid h_{1:T}, \theta) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi \exp(h_t)}} \exp\left(-\frac{r_t^2}{2 \exp(h_t)}\right). \quad (68)$$

Equivalently,

$$p(r_t \mid h_t, \theta) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}h_t - \frac{1}{2}r_t^2 \exp(-h_t)\right). \quad (69)$$

The second term in (65) describes the probability law of the latent volatility process. As presented in the previous chapter, under the autoregressive specification for log-volatility, this density can be factorized as

$$p(h_{1:T} \mid \theta) = p(h_1 \mid \theta) \prod_{t=2}^T p(h_t \mid h_{t-1}, \theta). \quad (70)$$

This term captures the dynamic dependence of volatility over time. The third term, $p(\theta)$, is the prior distribution assigned to the model parameters.

The reason Bayesian inference fits well with stochastic volatility models is evident from Equation (65). As the volatility is hidden, one cannot conduct inference solely on the basis of the likelihood function of observed returns. The Bayesian method considers the unobserved process of volatility as another unknown and evaluates it along with the model parameters.

4.2 Joint inference on parameters and latent volatility

A key feature of Bayesian stochastic volatility analysis is that the parameters and latent states are jointly inferred.

The joint posterior in Equation (65) can be decomposed in two useful ways.

First,

$$p(\theta, h_{1:T} \mid r_{1:T}) = p(\theta \mid h_{1:T}, r_{1:T})p(h_{1:T} \mid r_{1:T}). \quad (71)$$

Second,

$$p(\theta, h_{1:T} \mid r_{1:T}) = p(h_{1:T} \mid \theta, r_{1:T})p(\theta \mid r_{1:T}). \quad (72)$$

One can observe that the posterior distribution for the parameters depends on the possible volatility paths, while the posterior distribution for the volatility path depends on the possible parameter values.

The marginal posterior distribution of the parameters is obtained by integrating out the latent path:

$$p(\theta \mid r_{1:T}) = \int p(\theta, h_{1:T} \mid r_{1:T}) dh_{1:T}. \quad (73)$$

Similarly, the marginal posterior distribution of the latent volatility at a single time point is obtained by integrating out the other states and the parameters:

$$p(h_t \mid r_{1:T}) = \int p(\theta, h_{1:T} \mid r_{1:T}) d\theta dh_{-t}, \quad (74)$$

where h_{-t} denotes all latent states except h_t . Both of these formulas involve high-dimensional integration.

In an SV model, after the returns are observed, h_t can be inferred in 2 different ways. The filtered distribution of log-volatility

$$p(h_t \mid r_{1:t}), \quad (75)$$

uses only the information available up to time t , and the smoothed distribution

$$p(h_t \mid r_{1:T}), \quad (76)$$

which uses the whole sample, including the future observations. Since $\{r_{t+1}, \dots, r_T\}$ contain information about the volatility level around time t , smoothed volatility is generally more informative. State-space methods, that were slightly touched in the previous chapter, provide the general language for this filtering and smoothing distinction (Durbin and Koopman, 2012 [6]),

while Bayesian SV uses it to quantify uncertainty about the hidden volatility path (Jacquier, Polson and Rossi, 1994 [13]; Shephard, 1996 [18]).

4.3 Prior specification in SV models

Prior specification is an essential part of Bayesian stochastic volatility modelling because the prior must respect the restrictions imposed by the latent volatility process (Jacquier, Polson and Rossi, 1994 [13]; Kim, Shephard and Chib, 1998 [14]). For the basic SV model, the parameter vector is

$$\theta = (\mu, \phi, \sigma_\eta^2), \quad (77)$$

Under a common independence assumption, the prior distribution can be factorized as

$$p(\theta) = p(\mu)p(\phi)p(\sigma_\eta^2). \quad (78)$$

This assumption is not required, but it is convenient because it separates the interpretation of the three model parameters (Jacquier, Polson and Rossi, 1994 [13]; Chib, Nardari and Shephard, 2002 [4]).

The long-run level μ is unrestricted on the real line. A standard choice is therefore a normal prior,

$$\mu \sim N(m_0, s_0^2), \quad (79)$$

where m_0 represents the prior mean and s_0^2 represents prior uncertainty about the level of log-volatility.

The persistence parameter ϕ must satisfy the stationarity condition

$$|\phi| < 1. \quad (80)$$

Hence the prior for ϕ should assign probability only to the stationary interval $(-1, 1)$ when the model is intended to describe a stationary log-volatility process.

One way to enforce this restriction is to define a reparametrization by using an unrestricted

auxiliary parameter $x \in \mathbb{R}$ and set

$$\phi = \frac{\exp(x) - 1}{\exp(x) + 1} = \tanh\left(\frac{x}{2}\right), \quad (81)$$

Since $\exp(x) > 0$ for all $x \in \mathbb{R}$, this transformation maps the real line into the stationary interval $-1 < \phi < 1$.

In financial applications, where volatility is usually highly persistent, the prior for ϕ is often chosen to place substantial mass near one while still respecting $|\phi| < 1$.

The innovation variance must satisfy $\sigma_\eta^2 > 0$. A traditional prior choice is the inverse-gamma distribution,

$$\sigma_\eta^2 \sim IG(a_0, b_0), \quad (82)$$

where $a_0 > 0$ and $b_0 > 0$. This prior enforces positivity and controls the prior probability assigned to very small or very large values of the volatility-of-volatility parameter.

Finally, the initial latent state density $p(h_1 | \theta)$ is given by (56). Thus the full prior contribution for the latent path can be initialized consistently with the unconditional distribution of the log-volatility process.

Combining these components, a common prior specification for the basic SV model is

$$p(\theta, h_1) = p(\mu)p(\phi)p(\sigma_\eta^2)p(h_1 | \theta), \quad (83)$$

The role of the prior is therefore not only to introduce information before observing the sample, but also to ensure that the model respects the mathematical constraints of stochastic volatility. In practice, weakly informative priors are often used to restrict implausible parameter regions without overwhelming the likelihood when the data are informative (Gelman *et al.*, 2013 [10]).

4.4 Posterior distribution and latent-state uncertainty

From the joint posterior distribution $p(\theta, h_{1:T} \mid r_{1:T})$, one can obtain marginal posterior distributions for the individual parameters:

$$p(\mu \mid r_{1:T}), \quad p(\phi \mid r_{1:T}), \quad p(\sigma_\eta^2 \mid r_{1:T}). \quad (84)$$

These marginal posterior distributions quantify parameter uncertainty conditional on the observed return sample.

The same idea applies to the latent volatility states. For each $t = 1, \dots, T$, the smoothed posterior distribution of log-volatility is

$$p(h_t \mid r_{1:T}). \quad (85)$$

Since the conditional variance is

$$\sigma_t^2 = \exp(h_t), \quad (86)$$

posterior inference can be expressed either in terms of log-volatility h_t , variance σ_t^2 , or volatility $\sigma_t = \exp(h_t/2)$. One possible pointwise summary of the smoothed posterior distribution is the posterior mean,

$$\hat{h}_t = \mathbb{E}(h_t \mid r_{1:T}), \quad (87)$$

which represents the conditional expectation of log-volatility at time t given the full return sample.

The posterior predictive distribution becomes relevant in finance applications because of its role in building risk measures like Value-at-Risk. Two models might generate similar posterior mean volatility paths, but have very different credible bands. A wider credible band indicate a higher level of uncertainty about the risk involved, despite having similar central estimates. Such an insight holds importance in a financial setting because risk management depends on both expected volatility and its uncertainty about volatility along with the future returns distribution (Kim et al., 1998 [14]; Andersen et al., 2006 [1]; Tsay, 2010 [22]).

4.5 Analytical Intractability of the Posterior

The posterior distribution in Equation (65) is not analytically tractable in the standard stochastic volatility model. In this context, analytical intractability means that the posterior can be written up to proportionality, but it cannot usually be normalized, marginalized, or reduced to a known closed-form distribution.

Recalling the joint posterior of the parameters and latent volatility path,

$$p(\theta, h_{1:T} \mid r_{1:T}) \propto p(r_{1:T} \mid h_{1:T}, \theta) p(h_{1:T} \mid \theta) p(\theta), \quad (88)$$

the difficulty appears when one wants to obtain marginal posterior distributions. For example, the marginal posterior of the parameters requires integrating out the entire latent path:

$$p(\theta \mid r_{1:T}) = \int_{\mathbb{R}^T} p(\theta, h_{1:T} \mid r_{1:T}) dh_{1:T}. \quad (89)$$

This is a T -dimensional integral, where T is the length of the observed time series.

The main mathematical obstruction comes from the fact that the likelihood is non-Gaussian when viewed as a function of the latent log-volatility. Starting from the standard SV setting

$$\begin{cases} \text{observation equation: } r_t \mid h_t \sim N(0, e^{h_t}), \\ \text{state equation: } h_t \mid h_{t-1}, \theta \sim N(\mu + \phi(h_{t-1} - \mu), \sigma_\eta^2), \end{cases} \quad (90)$$

the conditional density of one observation is

$$p(r_t \mid h_t) = \frac{1}{\sqrt{2\pi e^{h_t}}} \exp\left(-\frac{r_t^2}{2e^{h_t}}\right). \quad (91)$$

Then its log-density as a function of h_t is

$$\log p(r_t \mid h_t) = -\frac{1}{2} \log(2\pi) - \frac{h_t}{2} - \frac{r_t^2}{2} e^{-h_t}. \quad (92)$$

Although $r_t \mid h_t$ is Gaussian as a distribution for the observed return r_t , Equation (92) shows

that the likelihood is not Gaussian as a function of h_t . A Gaussian density in h_t would have a log-density that is quadratic in h_t . Here, however, the term e^{-h_t} makes the log-density nonlinear and non-quadratic.

This matters because Bayesian updating combines the likelihood with the prior information implied by the state equation. The Gaussian transition density of the latent state is

$$p(h_t | h_{t-1}, \theta) \propto \exp \left(-\frac{[h_t - \mu - \phi(h_{t-1} - \mu)]^2}{2\sigma_\eta^2} \right). \quad (93)$$

If the likelihood were also Gaussian in h_t , multiplying it by this Gaussian transition density would produce a Gaussian posterior density. Equivalently, the log-posterior would remain quadratic in h_t , allowing closed-form expressions for posterior moments.

In the SV model, however, the conditional posterior of h_t is proportional to

$$p(h_t | r_t, h_{t-1}, \theta) \propto \exp \left(-\frac{h_t}{2} - \frac{r_t^2}{2} e^{-h_t} - \frac{[h_t - \mu - \phi(h_{t-1} - \mu)]^2}{2\sigma_\eta^2} \right). \quad (94)$$

The exponent in Equation (94) is not quadratic in h_t , due to the nonlinear term e^{-h_t} . Therefore, the conditional posterior cannot be reduced to a known Gaussian form with explicit mean and variance. This represents a loss of conjugacy between the Gaussian state equation and the observation density.

In the full sample, the same issue appears for every $t = 1, \dots, T$. The likelihood contributes nonlinear terms of the form e^{-h_t} throughout the latent path, so the joint posterior of $h_{1:T}$ is not multivariate Gaussian and the marginalization integral in Equation (89) has no closed-form solution in general.

For this reason, Bayesian inference in SV models relies on approximation and simulation methods. Posterior quantities are approximated using empirical averages obtained from methods such as data augmentation, MCMC methods, Gibbs sampling, and Metropolis–Hastings which will be presented in the following chapter (Jacquier, Polson and Rossi, 1994 [13]; Kim, Shephard and Chib, 1998 [14]; Chib, Nardari and Shephard, 2002 [4]).

5 Computational approaches to Bayesian SV Estimation

The basic SV model considered throughout the chapter is

$$r_t = \exp(h_t/2)\varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad (95)$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2), \quad (96)$$

with $|\phi| < 1$. Conditional on h_t , the return density is

$$p(r_t | h_t, \theta) = \frac{1}{\sqrt{2\pi \exp(h_t)}} \exp\left(-\frac{r_t^2}{2 \exp(h_t)}\right). \quad (97)$$

The computational difficulty, as presented in the previous chapter, is caused by the fact that $h_{1:T}$ is unobserved, high-dimensional, and enters the return density nonlinearly through $\exp(h_t)$.

5.1 Why computation is necessary

The previous chapter showed that the posterior distribution of the SV model is analytically intractable because inference involves both the static parameters and the latent volatility path. In computational terms, the central difficulty is that the likelihood of the observed returns requires integrating out the unobserved states:

$$p(r_{1:T} | \theta) = \int_{\mathbb{R}^T} p(r_{1:T}, h_{1:T} | \theta) dh_{1:T}. \quad (98)$$

Using the state-space factorization, the joint density can be written as

$$p(r_{1:T}, h_{1:T} | \theta) = p(h_1 | \theta) \prod_{t=2}^T p(h_t | h_{t-1}, \theta) \prod_{t=1}^T p(r_t | h_t, \theta). \quad (99)$$

The integral in Equation (98) is T -dimensional and generally unavailable in closed form.

Therefore, Bayesian inference for SV models relies on computational methods that avoid direct analytical integration over $h_{1:T}$. The key idea is to treat the latent volatility path as part of

the posterior object and to approximate posterior quantities through simulation.

5.2 Data augmentation and latent-volatility sampling

The idea of data augmentation, introduced by Tanner and Wong (1987) [19], is to enlarge the observed-data problem by treating unobserved quantities as latent data. In their notation, the observed data y are augmented with latent data z , so that the augmented data $x = (y, z)$ are easier to analyze than the observed data alone. Recall that in the SV setting, the observed data are the returns $r_{1:T}$, while the latent data are the unobserved log-volatility states $h_{1:T}$.

If the latent path $h_{1:T}$ were known, inference about the parameters $\theta = (\mu, \phi, \sigma_\eta^2)$ would be simpler, because the state equation would reduce to an observed Gaussian autoregression:

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \quad (100)$$

The marginal posterior distribution of the parameters is

$$p(\theta \mid r_{1:T}) = \int p(\theta, h_{1:T} \mid r_{1:T}) dh_{1:T} \quad (101)$$

which is difficult to evaluate analytically. Data augmentation avoids this direct marginalization by working instead with the joint posterior distribution $p(\theta, h_{1:T} \mid r_{1:T})$. This is studied through the augmented space containing both parameters and latent volatilities.

Following the logic of Tanner and Wong (1987) [19], the computation alternates between two conceptual steps. The first is an imputation step, in which latent data are drawn conditional on the observed data and the current parameter values. In the SV model this corresponds to sampling the latent volatility path:

$$h_{1:T}^{(m)} \sim p(h_{1:T} \mid \theta^{(m-1)}, r_{1:T}). \quad (102)$$

The second is a posterior step, in which the parameters are updated conditional on the

augmented data:

$$\theta^{(m)} \sim p(\theta \mid h_{1:T}^{(m)}, r_{1:T}). \quad (103)$$

In this way, the algorithm replaces the problem of analytically integrating over all hidden volatilities with the problem of repeatedly sampling hidden volatilities and parameters from their conditional posteriors.

The augmented complete-data log-density makes the structure of the problem explicit. Ignoring constants that do not depend on θ or $h_{1:T}$, the observation contribution is

$$\log p(r_{1:T} \mid h_{1:T}, \theta) = -\frac{1}{2} \sum_{t=1}^T (h_t + r_t^2 \exp(-h_t)) + C. \quad (104)$$

The state contribution coming from the Gaussian AR(1) state equation, can be written as conditional on the initial state h_1 due to Markov Property, and is expressed as

$$\log p(h_{2:T} \mid h_1, \theta) = -\frac{T-1}{2} \log(\sigma_\eta^2) - \frac{1}{2\sigma_\eta^2} \sum_{t=2}^T [h_t - \mu - \phi(h_{t-1} - \mu)]^2 + C. \quad (105)$$

Equations (104) and (105) show that the latent volatility path is tied to the data and to the parameters in two different ways. The observation density links h_t to the magnitude of returns: large values of $|r_t|$ tend to favour larger values of h_t , since higher log-volatility increases the conditional variance of returns. The state density, on the other hand, imposes temporal smoothness through the AR(1) structure and penalizes paths that fluctuate too irregularly, unless the innovation variance σ_η^2 is large.

This is precisely what makes data augmentation computationally appealing for use with Bayesian SV models. This methodology does not eliminate the latent volatility process from the posterior, but rather incorporates it into the inferential object of interest. Posterior inference is then based on simulated draws

$$\left(\theta^{(m)}, h_{1:T}^{(m)} \right), \quad m = 1, \dots, M,$$

from the joint posterior distribution. Marginal posterior summaries are obtained by ignoring

the components that are not of immediate interest. For example, the posterior mean of a parameter can be approximated by

$$\mathbb{E}(\theta_j \mid r_{1:T}) \approx \frac{1}{M} \sum_{m=1}^M \theta_j^{(m)}, \quad (106)$$

while the posterior mean of the latent log-volatility at time t can be approximated by

$$\mathbb{E}(h_t \mid r_{1:T}) \approx \frac{1}{M} \sum_{m=1}^M h_t^{(m)}. \quad (107)$$

5.3 MCMC methods

Before introducing Markov Chain Monte Carlo, it is useful to recall the basic definition of a Markov chain. Let

$$X^{(0)}, X^{(1)}, X^{(2)}, \dots$$

be a sequence of random variables taking values in a state space $\mathcal{X}$. This sequence is a Markov chain if the distribution of the next state depends only on the current state and not on the previous history:

$$\mathbb{P}(X^{(m+1)} \in A \mid X^{(m)}, X^{(m-1)}, \dots, X^{(0)}) = \mathbb{P}(X^{(m+1)} \in A \mid X^{(m)}), \quad (108)$$

for any measurable set $A \subseteq \mathcal{X}$. The transition mechanism of the chain is described by a transition kernel $K(x, A)$, defined as

$$K(x, A) = \mathbb{P}(X^{(m+1)} \in A \mid X^{(m)} = x). \quad (109)$$

A probability distribution π is called stationary, or invariant, for the Markov chain if

$$\pi(A) = \int_{\mathcal{X}} K(x, A) \pi(dx), \quad A \subseteq \mathcal{X}. \quad (110)$$

This means that if $X^{(m)} \sim \pi$, then also $X^{(m+1)} \sim \pi$. In other words, once the chain has

reached the stationary distribution, applying one more transition step does not change its distribution.

Coming back to the main topic of the subchapter, Markov chain Monte Carlo methods use this idea for posterior simulation. Instead of drawing independent samples directly from the posterior distribution, which is usually impossible in complex Bayesian models, MCMC constructs a Markov chain whose stationary distribution is the posterior distribution of interest. For Bayesian stochastic volatility, the target distribution is

$$\pi(x) = p(\theta, h_{1:T} \mid r_{1:T}), \quad \text{where } x = (\theta, h_{1:T}). \quad (111)$$

The aim is therefore to construct a transition kernel $K(x, A)$ such that the posterior distribution satisfies the invariance condition in Equation (110).

Since direct independent sampling from $\pi(x)$ is generally impossible, MCMC generates dependent draws

$$x^{(1)}, x^{(2)}, \dots, x^{(M)}, \quad x^{(m)} = (\theta^{(m)}, h_{1:T}^{(m)}), \quad (112)$$

using a transition rule that leaves π invariant. Under suitable regularity conditions (irreducibility and aperiodicity), the distribution of the Markov chain converges to π . After discarding an initial burn-in period, the retained draws are treated as an approximate dependent sample from the posterior distribution. The foundations of this simulation approach are given by the Metropolis algorithm and its generalization by Hastings (Metropolis *et al.*, 1953 [16]; Hastings, 1970 [12]).

Because the Markov chain is usually initialized at an arbitrary starting value, the early draws may still be strongly influenced by this initialization and may not yet represent the target posterior distribution well. The initial part of the chain is therefore commonly discarded, that part being called the burn-in period. If B denotes the number of burn-in iterations, then only the draws

$$x^{(B+1)}, \dots, x^{(M)}$$

are used for posterior summaries.

Posterior expectations can then be approximated by Monte Carlo averages. Formally, for any function $g(\theta, h_{1:T})$, the posterior expectation is

$$\mathbb{E}[g(\theta, h_{1:T}) \mid r_{1:T}] = \int g(\theta, h_{1:T}) p(\theta, h_{1:T} \mid r_{1:T}) d\theta dh_{1:T}. \quad (113)$$

MCMC replaces this posterior integral with an empirical average over the retained draws:

$$\mathbb{E}[g(\theta, h_{1:T}) \mid r_{1:T}] \approx \frac{1}{M-B} \sum_{m=B+1}^M g(\theta^{(m)}, h_{1:T}^{(m)}), \quad (114)$$

For example, taking $g(\theta, h_{1:T}) = \phi$ gives the posterior mean of the persistence parameter:

$$\mathbb{E}(\phi \mid r_{1:T}) \approx \frac{1}{M-B} \sum_{m=B+1}^M \phi^{(m)}. \quad (115)$$

Similarly, taking $g(\theta, h_{1:T}) = \exp(h_t/2)$ gives the posterior mean of the latent conditional volatility at time t :

$$\mathbb{E}[\exp(h_t/2) \mid r_{1:T}] \approx \frac{1}{M-B} \sum_{m=B+1}^M \exp(h_t^{(m)}/2). \quad (116)$$

The relevance of MCMC for stochastic volatility is that it turns an analytically intractable posterior into a simulable object. The cost is that the draws are correlated, convergence must be assessed, and poor proposal choices may lead to slow mixing (Robert and Casella, 2004 [17]; Gelman et al., 2013 [10]).

5.4 Gibbs sampling and Metropolis-Hastings

This chapter explains the two main MCMC building blocks used in Bayesian stochastic volatility: Gibbs Sampling and Metropolis-Hastings. In practice, a sampler usually combines both, leading to Metropolis-Hastings within Gibbs algorithm.

5.4.1 Gibbs Sampling

Gibbs sampling is an MCMC method used when direct sampling from a joint posterior distribution is difficult. In the Bayesian SV setting, the target distribution is the joint posterior

$$p(\theta, h_{1:T} \mid r_{1:T}),$$

where $\theta = (\mu, \phi, \sigma_\eta^2)$. A proper blocking scheme separates the latent volatility path and the model parameters:

$$h_{1:T}, \quad \mu, \quad \phi, \quad \sigma_\eta^2. \quad (117)$$

The Gibbs sampler therefore cycles through the corresponding full conditional distributions. In this setting, these conditionals take the form

$$p(h_{1:T} \mid \mu, \phi, \sigma_\eta^2, r_{1:T}), \quad (118)$$

$$p(\mu \mid \phi, \sigma_\eta^2, h_{1:T}, r_{1:T}), \quad (119)$$

$$p(\phi \mid \mu, \sigma_\eta^2, h_{1:T}, r_{1:T}), \quad (120)$$

$$p(\sigma_\eta^2 \mid \mu, \phi, h_{1:T}, r_{1:T}). \quad (121)$$

In the SV model, observed returns depend on the latent volatility states, while the latent states depend on static parameters. Gibbs sampling exploits this conditional structure by updating one block at a time while conditioning on the current values of the remaining blocks.

The Gibbs sampler was introduced in the context of Bayesian image restoration by Geman and Geman (1984) [11], where local conditional distributions were used to generate a Markov chain over image configurations. Its relevance for Bayesian statistics was later emphasized by Gelfand and Smith (1990) [9], who showed how sampling from full conditional distributions could be used to approximate posterior marginal distributions in a broad class of models.

Some full conditional distributions in the SV model can be derived from the Gaussian AR(1)

state equation. Define the state residuals

$$e_t(\mu, \phi) = h_t - \mu - \phi(h_{t-1} - \mu), \quad t = 2, \dots, T. \quad (122)$$

Throughout this section, the inverse-gamma distribution (IG) is parameterized as

$$x \sim IG(a, b) \iff p(x) = \frac{b^a}{\Gamma(a)} x^{-(a+1)} \exp\left(-\frac{b}{x}\right), \quad x > 0. \quad (123)$$

So if an inverse-gamma prior is used,

$$\sigma_\eta^2 \sim IG(a_0, b_0), \quad (124)$$

then, conditional on μ , ϕ , and $h_{1:T}$, the posterior distribution of σ_η^2 is also inverse-gamma:

$$\sigma_\eta^2 \mid \mu, \phi, h_{1:T} \sim IG\left(a_0 + \frac{T-1}{2}, b_0 + \frac{1}{2} \sum_{t=2}^T e_t(\mu, \phi)^2\right), \quad (125)$$

up to the treatment of the initial-state density h_1 .

A similar conjugate update can be obtained for μ under a normal prior. Rearranging the state equation gives

$$h_t - \phi h_{t-1} = (1 - \phi)\mu + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \quad (126)$$

Define

$$y_t = h_t - \phi h_{t-1}, \quad x_\phi = 1 - \phi. \quad (127)$$

Then, conditional on ϕ , σ_η^2 , and $h_{1:T}$, the state equation can be written as

$$y_t = x_\phi \mu + \eta_t, \quad t = 2, \dots, T. \quad (128)$$

This is a Gaussian regression model with known error variance σ_η^2 and unknown coefficient μ .

Assume the normal prior

$$\mu \sim N(m_0, s_0^2). \quad (129)$$

Ignoring the initial-state term for simplicity, the conditional posterior density of μ is proportional to

$$p(\mu \mid \phi, \sigma_\eta^2, h_{1:T}) \propto \exp \left(-\frac{1}{2s_0^2}(\mu - m_0)^2 - \frac{1}{2\sigma_\eta^2} \sum_{t=2}^T (y_t - x_\phi \mu)^2 \right). \quad (130)$$

Completing the square in μ gives a normal full conditional distribution:

$$\mu \mid \phi, \sigma_\eta^2, h_{1:T} \sim N(m_\mu, V_\mu), \quad (131)$$

where

$$V_\mu = \left(\frac{1}{s_0^2} + \frac{(T-1)x_\phi^2}{\sigma_\eta^2} \right)^{-1}, \quad (132)$$

$$m_\mu = V_\mu \left(\frac{m_0}{s_0^2} + \frac{x_\phi}{\sigma_\eta^2} \sum_{t=2}^T y_t \right). \quad (133)$$

Substituting back $x_\phi = 1 - \phi$ and $y_t = h_t - \phi h_{t-1}$ gives

$$V_\mu = \left(\frac{1}{s_0^2} + \frac{(T-1)(1-\phi)^2}{\sigma_\eta^2} \right)^{-1}, \quad (134)$$

$$m_\mu = V_\mu \left(\frac{m_0}{s_0^2} + \frac{1-\phi}{\sigma_\eta^2} \sum_{t=2}^T (h_t - \phi h_{t-1}) \right). \quad (135)$$

Conditional on the latent volatility path and the remaining parameters, μ can be sampled directly from this normal distribution given by (131).

A problem arises due to the fact that some conditionals may not necessarily follow standard distributions. The conditional distribution of ϕ can be non-standard, especially when we take into account the stationarity restriction $|\phi| < 1$ or the density of the initial state. Furthermore, the conditional distribution of latent process $h_{1:T}$ can also be non-normal. Thus, a pure Gibbs sampling algorithm is generally not sufficient for the standard SV model.

5.4.2 Metropolis–Hastings

Let $\pi(x)$ denote the target density and let $q(x^* | x)$ be a proposal density. Starting from the current value x , the algorithm proposes a candidate value

$$x^* \sim q(x^* | x).$$

The candidate is accepted with probability

$$\alpha(x, x^*) = \min \left\{ 1, \frac{\pi(x^*)q(x | x^*)}{\pi(x)q(x^* | x)} \right\}. \quad (136)$$

If the proposal is accepted, the next state is set equal to x^* . If it is rejected, the chain remains at the current value x . This accept-reject mechanism defines a Markov transition whose invariant distribution is π .

The reason the method is useful for Bayesian posterior simulation is that the normalizing constant of the target density cancels in the acceptance ratio. Suppose that the target density can be written as

$$\pi(x) = \frac{\tilde{\pi}(x)}{C},$$

where $\tilde{\pi}(x)$ is known and C is an unknown normalizing constant. Then

$$\frac{\pi(x^*)}{\pi(x)} = \frac{\tilde{\pi}(x^*)/C}{\tilde{\pi}(x)/C} = \frac{\tilde{\pi}(x^*)}{\tilde{\pi}(x)}. \quad (137)$$

Therefore the method can be applied to posterior densities that are known only up to proportionality (Hastings, 1970 [12]; Robert and Casella, 2004 [17]).

In stochastic volatility models, x may be a scalar parameter such as ϕ , a single latent state h_t , a block of states, or the entire latent path. To see why Metropolis–Hastings is useful, consider an interior latent state h_t , where $2 \leq t \leq T - 1$. Conditional on the neighboring states, the parameters, and the return r_t , the full conditional distribution of h_t satisfies

$$p(h_t | h_{t-1}, h_{t+1}, \mu, \phi, \sigma_\eta^2, r_t) \propto p(r_t | h_t) p(h_t | h_{t-1}, \mu, \phi, \sigma_\eta^2) p(h_{t+1} | h_t, \mu, \phi, \sigma_\eta^2). \quad (138)$$

Using the SV observation equation and the Gaussian AR(1) state equation, the corresponding log-target is

$$\begin{aligned} \log p(h_t \mid h_{t-1}, h_{t+1}, \mu, \phi, \sigma_\eta^2, r_t) = & -\frac{1}{2} (h_t + r_t^2 \exp(-h_t)) \\ & -\frac{1}{2\sigma_\eta^2} [h_t - \mu - \phi(h_{t-1} - \mu)]^2 \\ & -\frac{1}{2\sigma_\eta^2} [h_{t+1} - \mu - \phi(h_t - \mu)]^2 + C, \end{aligned} \quad (139)$$

where C is a constant term that collects all the terms that do not depend on h_t .

The nonlinear term $r_t^2 \exp(-h_t)$ means that this conditional density is not Gaussian and cannot be sampled from directly using a standard Gibbs update. A Metropolis–Hastings step can therefore be used to update h_t .

5.4.3 Metropolis–Hastings-within-Gibbs for Stochastic Volatility

In a pure Gibbs sampler, each block is sampled directly from its full conditional distribution. The problem is that some full conditional distributions are not available in a standard form in many Bayesian models. The solution is to update the conjugate blocks by direct Gibbs sampling and to update the non-standard blocks using Metropolis–Hastings steps.

The resulting Markov chain still targets the joint posterior distribution

$$p(\theta, h_{1:T} \mid r_{1:T}),$$

provided that each block update leaves the corresponding full conditional distribution invariant. This condition is automatically satisfied by direct Gibbs updates and is satisfied by Metropolis–Hastings updates when the acceptance probability is constructed with the appropriate conditional target.

The following pseudocode summarizes a generic Metropolis–Hastings-within-Gibbs sampler for Bayesian stochastic volatility. (Gelfand and Smith, 1990 [9]; Tierney, 1994 [21]; Jacquier, Polson and Rossi, 1994 [13]; Kim, Shephard and Chib, 1998 [14]).

Algorithm 1 Metropolis–Hastings-within-Gibbs sampler for Bayesian stochastic volatility

Require: Observed returns $r_{1:T}$, initial values $\theta^{(0)}$ and $h_{1:T}^{(0)}$, number of iterations M

Ensure: Approximate posterior sample from $p(\theta, h_{1:T} \mid r_{1:T})$

1: **for** $m = 1, \dots, M$ **do**

2: Update the latent volatility path $h_{1:T}^{(m)}$ using a transition step targeting

$$p(h_{1:T} \mid \theta^{(m-1)}, r_{1:T}).$$

If direct sampling is not available, use one or more Metropolis–Hastings updates. For a generic latent-volatility block b , propose

$$h_b^* \sim q_b(h_b^* \mid h_b^{(m-1)}),$$

and accept it with probability

$$\alpha_b = \min \left\{ 1, \frac{p(h_b^* \mid h_{-b}, \theta^{(m-1)}, r_{1:T}) q_b(h_b^{(m-1)} \mid h_b^*)}{p(h_b^{(m-1)} \mid h_{-b}, \theta^{(m-1)}, r_{1:T}) q_b(h_b^* \mid h_b^{(m-1)})} \right\}.$$

3: Update $\mu^{(m)}$ from its full conditional distribution, if available in closed form; otherwise use a Metropolis–Hastings step.

4: Update $\phi^{(m)}$ from its full conditional distribution if available; otherwise propose

$$\phi^* \sim q_\phi(\phi^* \mid \phi^{(m-1)})$$

and accept it with the Metropolis–Hastings probability targeting

$$p(\phi \mid \mu^{(m)}, \sigma_\eta^{2(m-1)}, h_{1:T}^{(m)}, r_{1:T}),$$

especially when the stationarity restriction $|\phi| < 1$ is imposed.

5: Update $\sigma_\eta^{2(m)}$ from its full conditional distribution, such as the inverse-gamma distribution in Equation (125), when the prior specification allows it.

6: Store $(\theta^{(m)}, h_{1:T}^{(m)})$.

7: **end for**

8: Discard burn-in draws and use the retained draws to compute posterior summaries.

6 Bayesian SV and Classical Conditional Volatility Models

The theoretical sections introduced two distinct ways of formalising time-varying volatility. The ARCH/GARCH models were presented as conditional heteroskedasticity models in which the variance is updated by an observation-driven recursion. Stochastic volatility models were later defined as state-space models in which volatility is an unobserved stochastic process.

6.1 Different interpretations of volatility

The central difference between GARCH and stochastic volatility models is not only the functional form of the variance equation, but the probabilistic status assigned to volatility. In a GARCH model, the conditional variance is an observation-driven process. Once the parameters and the past information set $\mathcal{F}_{t-1}$ are given, the conditional variance is measurable with respect to $\mathcal{F}_{t-1}$:

$$\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1}). \quad (140)$$

By contrast, stochastic volatility models are parameter-driven models. The volatility process is governed by an additional latent state equation. In the standard SV model, the conditional variance satisfies

$$\text{Var}(r_t \mid h_t) = \exp(h_t). \quad (141)$$

Unlike σ_t^2 in GARCH, h_t is not measurable with respect to the observed return history $\mathcal{F}_{t-1}$. Instead, it has its own conditional distribution given past latent states and parameters. Thus, even when the parameters are known, the volatility path is not directly observed and must be inferred probabilistically.

This distinction can be expressed as

$$\text{GARCH:} \quad \sigma_t^2 = g(r_{t-1}, \sigma_{t-1}^2; \theta), \quad (142)$$

whereas

$$\text{SV:} \quad h_t \sim p(h_t \mid h_{t-1}, \theta), \quad r_t \sim p(r_t \mid h_t, \theta). \quad (143)$$

The first formulation updates volatility deterministically as a function of past observations, while the second treats volatility as a hidden stochastic state that generates the observed return.

From an inferential perspective, in GARCH, likelihood evaluation is based on a deterministic recursion for σ_t^2 . On the other hand, in SV models, the likelihood requires integrating over the latent volatility path:

$$p(r_{1:T} | \theta) = \int p(r_{1:T}, h_{1:T} | \theta) dh_{1:T}. \quad (144)$$

Thus, the difference between the two model classes is directly connected to the computational difficulty of Bayesian SV inference. GARCH requires estimation of parameters governing an observed-history recursion, while SV requires inference on both parameters and an unobserved volatility trajectory.

Economically, conditional volatility has a mechanical relation with the past return shocks through the recursiveness of conditional variance. In SV models, volatility is treated as a latent factor of risk which persists over time. It is thus clear that volatility clustering results from different causes: while GARCH models rely on the fact that big shocks directly affect future conditional variances, SV models rely on the persistence of the latent volatility state for generating periods of high and low uncertainty in the market.

6.2 The contribution of Bayesian inference

The importance of Bayesian approach comes particularly from its ability to deal with volatility modelling within the SV setting. Instead of assuming that volatility is determined by some fitted deterministic sequence, Bayesian analysis proceeds under the assumption that inference about volatility will be conducted in an augmented space including static parameters and the hidden states. For

$$\theta = (\mu, \phi, \sigma_\eta^2),$$

recall that the inferential target is the joint posterior distribution

$$p(\theta, h_{1:T} | r_{1:T}) \propto p(r_{1:T} | h_{1:T}, \theta) p(h_{1:T} | \theta) p(\theta). \quad (145)$$

The most significant difference from classical estimation approach is that the volatility path becomes part of the inferential object. Bayesian inference produces a probability distribution over both θ and $h_{1:T}$. Marginal posterior distributions are obtained by integration. For example,

$$p(\theta \mid r_{1:T}) = \int p(\theta, h_{1:T} \mid r_{1:T}) dh_{1:T}, \quad (146)$$

while the smoothed posterior distribution of a latent state is

$$p(h_t \mid r_{1:T}) = \int p(\theta, h_{1:T} \mid r_{1:T}) d\theta dh_{-t}, \quad (147)$$

where h_{-t} denotes the latent volatility path excluding h_t . Therefore, the uncertainty about the volatility state at each date is represented by a full posterior distribution.

This distinction is especially relevant for persistent volatility processes. When the persistence parameter $\phi \approx 1$, small differences in its posterior distribution can imply different conclusions about the duration of high-volatility periods. A point estimate of ϕ cannot fully represent this uncertainty.

To see the advantages of Bayesian Inference in prediction, consider the one-step-ahead predictive density obtained by integrating over parameter uncertainty, the current latent volatility state, and the future volatility innovation

$$p(r_{T+1} \mid r_{1:T}) = \int p(r_{T+1} \mid h_{T+1}, \theta) p(h_{T+1} \mid h_T, \theta) p(\theta, h_{1:T} \mid r_{1:T}) dh_{T+1} dh_{1:T} d\theta. \quad (148)$$

The predictive density is averaged over all posterior uncertainty rather than conditioned upon a single parameter estimate and a single volatility path estimate. This is one of the most relevant differences from the plug-in approach, where both the parameters and the volatility process are treated as constants once estimated.

A significant problem that this approach is facing is the requirement of considerable computational power. The evaluation of the likelihood needs integration over $h_{1:T}$, and the dimension of such integral increases with the sample size. As a result, Bayesian SV relies on simulation-based methods such as data augmentation, Gibbs sampling, Metropolis-Hastings updates, and other

Monte Carlo methods to solve this impediment (Robert and Casella, 2004 [17]; Chib, Nardari and Shephard, 2002 [4]).

6.3 Strengths and weaknesses of Bayesian SV

6.3.1 Strengths

The main advantage of Bayesian stochastic volatility is its unified treatment of latent volatility. Since h_t is not observed, representing it through a posterior distribution is more informative than reducing it to a single fitted path. A posterior mean provides one central estimate of volatility at time t , while a credible band constructed from the posterior quantiles of $p(h_t \mid r_{1:T})$ shows how uncertain this estimate is at that date. This is important because returns are often noisy. A large absolute return $|r_t|$ may provide evidence of high volatility, but it may also be an extreme observation generated during a period of moderate latent volatility.

A second strength is the treatment of parameter uncertainty. Each parameter of the SV state equation has direct theoretical meaning: μ controls the long-run log-volatility level, ϕ controls persistence, and σ_η^2 controls volatility-of-volatility. The marginal posterior distributions make uncertainty about each of these quantities explicit. In financial terms, this means uncertainty about the level, persistence and instability of future risk is propagated into forecasts instead of being suppressed by a plug-in estimate.

A third and final strength is the possibility for extensions. The basic Gaussian SV model can be extended in several directions, to heavy-tailed observation errors, leverage effects, jumps or multivariate volatility. These additions preserve the state-space logic and add parameters or latent variables to the posterior. Empirical work shows that such features are often important in financial returns, making the Bayesian SV framework well suited to the empirical regularities commonly observed in asset returns.

6.3.2 Weaknesses

The weaknesses, on the other hand, follow from the same latent structure. First, Bayesian SV is computationally more expensive than any classical conditional volatility model, due to the high-dimensionality and nonlinearity of the posterior.

Another issue relates to the requirement to perform diagnostic checks for the simulation process. Since Bayesian SV modelling usually relies on the MCMC algorithm for simulations, the accuracy of the obtained results hinges on whether the Markov chain explores the target posterior density properly. Otherwise, poor convergence can lead to the chain getting trapped in a local area of the posterior, resulting in biased estimates. Specifically, the choice of ϕ is crucial, especially when dealing with very persistent volatility processes. Another challenge involves the Metropolis–Hastings algorithm, where the quality of the selected proposal distribution plays a critical role in ensuring an informative output. The choice of the prior can affect the model too. Priors can enforce stationarity and positivity, but they may also affect inference when the data contain limited information about ϕ or σ_η^2 (Gelman *et al.*, 2013 [10]).

The importance of using Bayesian SV does not arise from the mere complexity of the model. Rather, it stems from the importance of the posterior distribution as a source of uncertainty related to both the latent process and forecasted uncertainty. GARCH still remains more appropriate when the task requires a fast and clear estimate of conditional variance.

7 Conclusions

7.1 Main Findings

This thesis analyzed how Bayesian inference addresses the estimation problem arising in stochastic volatility models. The most important finding is that Bayesian inference framework is appropriate to SV due to its intrinsically latent nature. Returns can be observed, but the dynamics of the conditional variance, which determine volatility, cannot. Hence, it follows that the posterior distribution is the right object for inference since it reflects our beliefs about both parameters and volatility itself in the light of observations.

The comparison to the family of ARCH/GARCH models gives more insight about the previous result. GARCH models are useful benchmarks from an efficiency point of view since they incorporate a deterministic recursion on the conditional variance and, therefore, represent a way of modelling volatility clustering. Bayesian inference is valuable exactly because it converts this latent state formulation into posterior distributions and predictive densities.

The main theoretical conclusion does not claim superiority of the Bayesian SV approach over GARCH in all applications. These two approaches tackle the problems differently. While GARCH deals with the recursion on conditional variance, Bayesian SV focuses on inference about latent volatility. The latter has multiple advantages which emerge when uncertainty about latent volatility and future risk is essential.

7.2 Limitations and future extensions

Limitations arise from the theoretical focus of this paper. It mainly looks at the basic univariate SV model, which is useful for explaining the main inference problem, but it omits several features that are often important in empirical finance, namely leverage effects, structural breaks, time-varying correlations, and multivariate covariance dynamics. It also does not provide a full empirical comparison across different assets or forecasting horizons of volatility. Because of this, the conclusions are more about the logic of the model and how inference works rather

than its out-of-sample forecasting performance.

This thesis could be extended in several directions. One possible extension would be to move to a multivariate SV framework, where the focus shifts to time-varying covariances and their implications for portfolio risk and asset allocation. A second possibility would be to bring in high-frequency information, such as realised volatility, to improve inference on the latent volatility process.

To conclude, Bayesian SV is relevant because it treats volatility as stochastic and uncertain. Its role is not to replace every conditional volatility model, but to provide a coherent probabilistic framework when latent risk and uncertainty quantification play an essential role in the modelling objective.

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Appendix

This appendix provides a simulation study to illustrate the Bayesian SV framework developed in the thesis. The source code is not reproduced in the appendix. Instead, the complete implementation of the simulation study is available in the following GitHub repository. Here we summarize the model setup, simulation design, and key outputs. All model parameters are fixed at their true data-generating values, so that the MCMC sampler is used only to sample the latent volatility path. (Full Bayesian estimation of the parameters would require sampling μ, ϕ, σ_η jointly with $h_{1:T}$, which is beyond the scope of this appendix.)

A Bayesian SV Simulation

A.1 Purpose of the Simulation

The purpose of this simulation is to provide an illustrative numerical example of the Bayesian stochastic volatility model described in the thesis. By constructing a controlled example with known parameters, we can verify the theoretical concepts introduced in the chapters on latent volatility, Bayesian inference, posterior uncertainty, and MCMC methods (specifically the Metropolis–Hastings algorithm). In particular, the simulation demonstrates how volatility clustering appears in the data and how it can be captured by Bayesian latent volatility estimation. The results are intended for illustrative verification of the theoretical framework, not for empirical forecasting.

A.2 Simulated Stochastic Volatility Model

We simulate data from the standard stochastic volatility model:

$$r_t = \exp\left(\frac{h_t}{2}\right) \varepsilon_t, \quad \varepsilon_t \sim N(0, 1),$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta \eta_t, \quad \eta_t \sim N(0, 1).$$

Here r_t is the observed return and h_t is the latent log-volatility. The parameter μ is the long-run mean of h_t , ϕ is the persistence parameter (with ϕ close to 1 indicating high persistence), and σ_η is the volatility-of-volatility controlling the innovation scale of h_t . We set the true parameters to

$$\mu = -10.0, \quad \phi = 0.98, \quad \sigma_\eta = 0.20,$$

and simulate a time series of length $T = 1000$. The initial h_1 is drawn from the stationary distribution of the AR(1) process. With $\phi = 0.98$, the latent volatility is highly persistent, and nonzero σ_η induces stochastic fluctuations around μ .

Quantity	Value
Sample size T	1000
Long-run log-volatility μ	-10.0
Persistence parameter ϕ	0.98
Volatility-of-volatility σ_η	0.20

Table 1: Simulation settings used in the appendix.

Figure 2 plots the simulated latent log-volatility path $\{h_t\}$. The series stays around $\mu = -10$ but shows persistent swings due to $\phi < 1$. Figure 3 shows the corresponding simulated returns r_t . Notice that large return magnitudes tend to cluster together, reflecting volatility clustering: periods when h_t is higher lead to larger $|r_t|$ values.

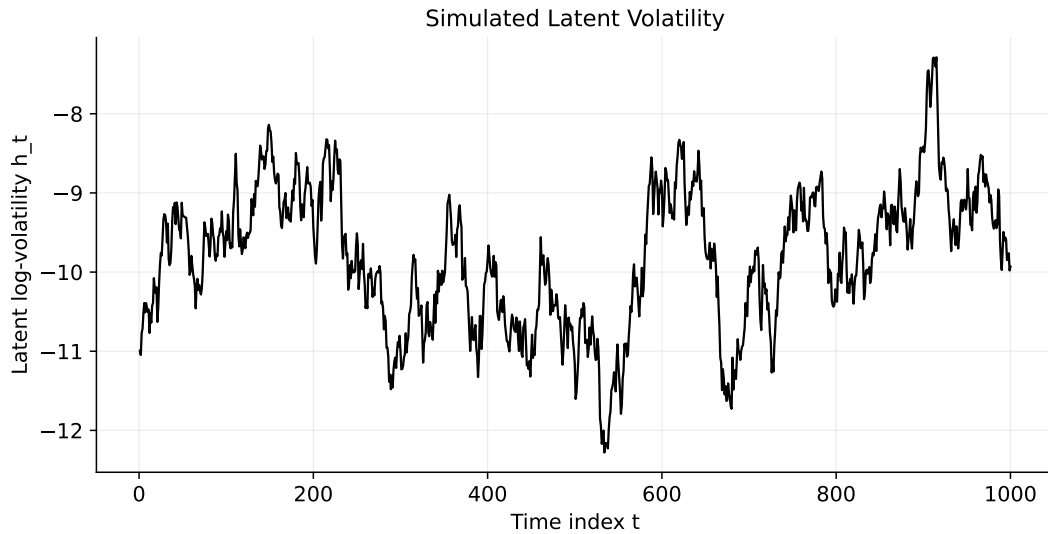


Figure 2: Simulated latent log-volatility path h_t for the SV model (parameters $\mu = -10$, $\phi = 0.98$, $\sigma_\eta = 0.20$).

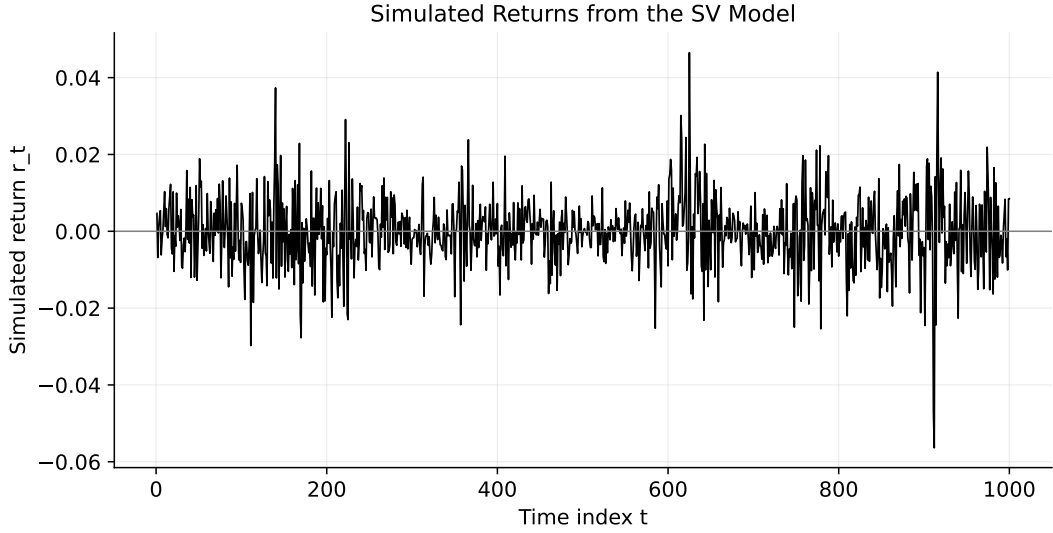


Figure 3: Simulated return series $r_t = \exp(h_t/2)\varepsilon_t$ from the SV model.

A.3 MCMC Sampling Procedure

We perform Bayesian inference on the latent path $h_{1:T}$ by a component-wise Metropolis–Hastings sampler, treating (μ, ϕ, σ_η) as known. The target posterior is $p(h_{1:T} \mid r_{1:T}, \mu, \phi, \sigma_\eta)$. The sampling algorithm proceeds as follows:

1. Initialize the latent path $h_{1:T}$ (e.g. by a simple function of the returns).
2. For iteration $i = 1, \dots, N_{\text{iter}}$ (with $N_{\text{iter}} = 12000$ draws):
 - (a) For each time $t = 1, \dots, T$, propose $h_t^* \sim N(h_t, \sigma_{\text{prop}}^2)$ with $\sigma_{\text{prop}} = 0.15$.
 - (b) Compute the Metropolis–Hastings acceptance probability using the local posterior contribution for h_t (involving the likelihood from r_t and the AR(1) prior terms with neighboring h_{t-1}, h_{t+1}).
 - (c) Accept h_t^* with this probability; otherwise retain the old h_t .
3. After the run, discard the first $B = 4000$ draws as burn-in and use the remaining 8000 draws as samples from the posterior of $h_{1:T}$.

In our setup we used $N_{\text{iter}} = 12000$, burn-in $B = 4000$, and proposal standard deviation 0.15. The overall acceptance rate was around 0.6908, suggesting that the proposal scale was not

excessively aggressive. The acceptance rates alone do not establish convergence or efficient mixing, so trace plots and autocorrelation diagnostics are also reported. (Recall that the SV parameters were fixed. A fully Bayesian estimation of $\{\mu, \phi, \sigma_\eta\}$ would require a more complex joint sampler, which is beyond the scope here.)

A.4 Posterior Credible Bands for Latent Volatility

Using the posterior samples of $h_{1:T}$, we compute the posterior mean and 95% credible interval at each t . Figure 4 shows the posterior mean path (blue line) and the 95% credible bands (shaded area), along with the true simulated h_t (black line). The posterior mean closely tracks the true latent volatility. For example, at $t = 500$ the posterior mean is about -11.3749 , the posterior standard deviation = 0.4055 , and the 95% credible interval width is about 1.4467 . The true h_t lies within the credible bands for most of the sample, indicating that posterior uncertainty is appropriately quantified given the fixed parameters.

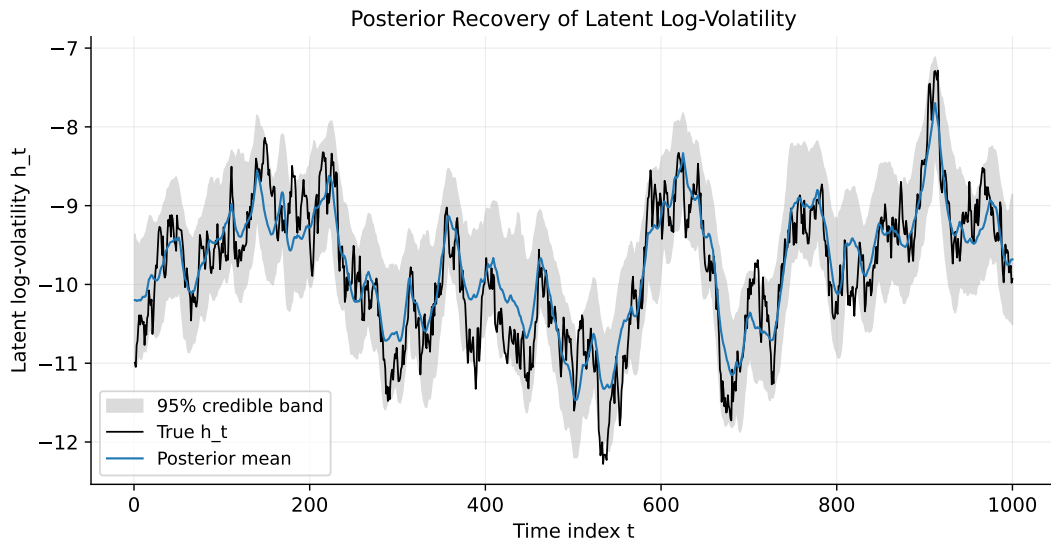


Figure 4: Posterior mean (blue line) and 95% credible bands (shaded) for the latent log-volatility h_t . The true simulated h_t (black line) is shown for reference.

A.5 MCMC Diagnostics

Figure 5 displays trace plots of the posterior draws, after burn-in, for several representative states ($h_{250}, h_{500}, h_{750}$). The chains appear to mix and explore the posterior without obvious divergence, though some autocorrelation is visible. Figure 6 shows the sample autocorrelation of the MCMC draws for h_{500} . As expected from a component-wise random-walk MH sampler, the draws display noticeable autocorrelation. The autocorrelation decreases with the lag, but rather slowly, reflecting the limited mixing efficiency of simple component-wise updates. We do not claim formal convergence because these diagnostics provide just a basic check that the sampler is behaving as intended.

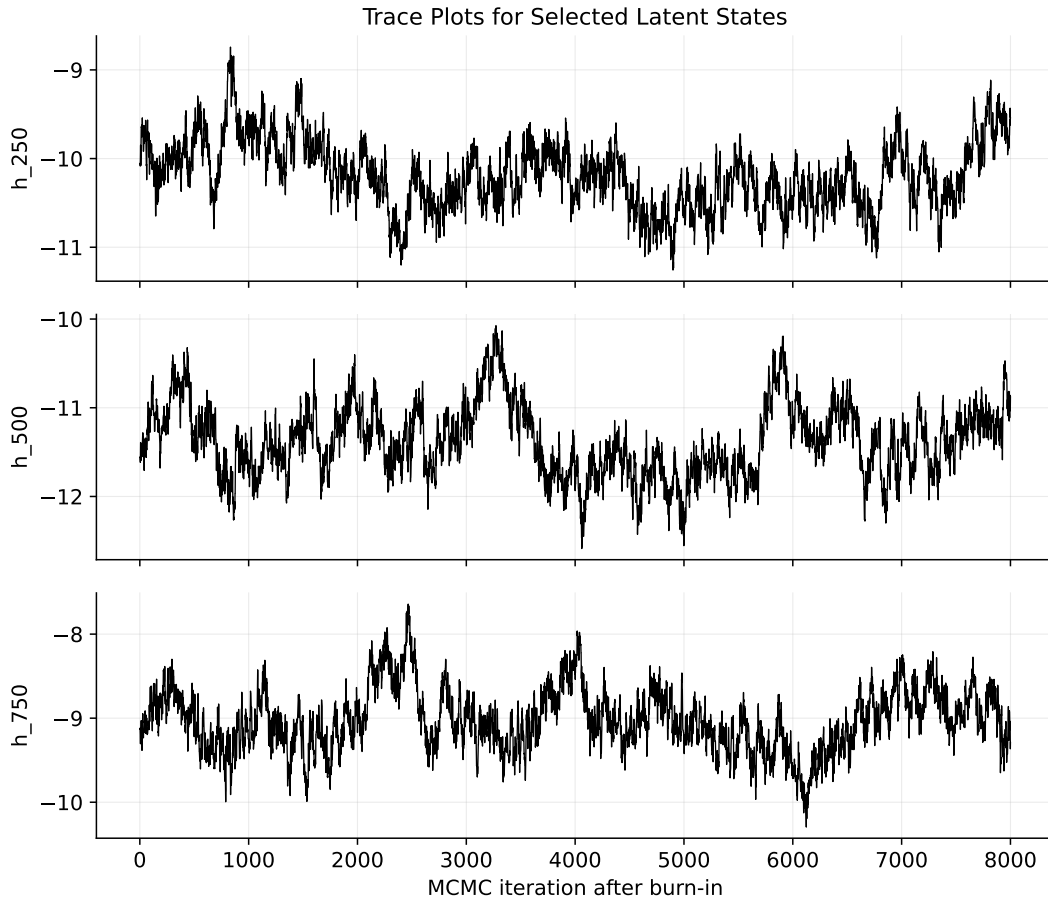


Figure 5: Trace plots for selected latent states ($h_{250}, h_{500}, h_{750}$) from the posterior draws (after burn-in).

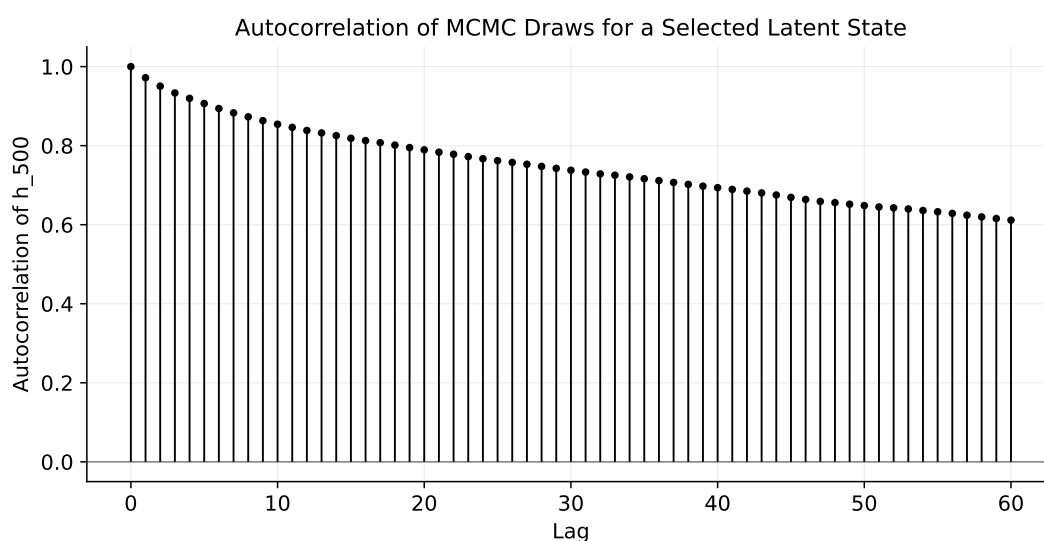


Figure 6: Sample autocorrelation (up to lag 60) of the MCMC draws for the latent state h_{500} .

A.6 Summary of the Appendix

The above simulation illustrates the key phenomena discussed in the thesis. The simulated returns in Figure 3 exhibit volatility clustering driven by the latent h_t process shown in Figure 2. The Bayesian sampler recovers the latent volatility path, and the posterior mean matches the true h_t well (Figure 4) with reasonable uncertainty bands. The MCMC diagnostics (Figures 5 and 6) provide a sanity check on the sampler's behavior. Note that throughout this appendix the SV model parameters were fixed at their data-generating values, a fully Bayesian treatment would also sample those parameters, which we leave for future work.